REPORT

OF

DATA SCIENCE MASTER VIRTUAL INTERNSHIP

Submitted

To

School of Computer Science and

Engineering IILM University, Greater

Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number : 2410030312

AYUSH KUMAR SINGH

Batch: 2026-27

CANDIDATE'S DECLARATION

I, Ayush Kumar Singh do hereby declare that the internship report titled **"Topic"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name:

AYUSH KUMAR SINGH

Roll No.: 2410030312

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature

Student Name:

AYUSH KUMAR SINGH

Roll No.: 2410030312

INTERNSHIP COMPLETION CERTIFICATE

Copy of the certificate issued for the completed internship



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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

The Data Science Master Virtual Internship was an 8-week learning programme completed during June-August 2026. The internship was designed to provide practical exposure to the data science workflow and to strengthen the learner's understanding of how data can be collected, prepared, analysed and interpreted for problem-solving.

The internship complemented the academic curriculum of B.Tech in Computer Science and Engineering by connecting theoretical concepts with practical data-oriented tasks. The virtual format also encouraged independent learning, regular practice and systematic completion of assigned learning activities.

4.2 Organization Profile

EduSkills is the internship platform identified on the certificate, with the certificate carrying the EduSkills branding and stating that the internship was supported by SIEMENS. The certificate also displays the Ministry of Education, AICTE and National Internship Portal branding. The programme was delivered in a virtual format and the certificate records successful completion of the Data Science Master Virtual Internship.

4.3 Problem Statement

Students studying computer science often learn data science concepts through individual subjects, but they also need structured practical exposure to understand how these concepts are applied in a complete workflow. The internship addressed this learning requirement by providing a focused virtual learning experience in data science and by encouraging the application of analytical concepts to practical tasks.

4.4 Project Objectives

The major objectives of the internship report and associated learning programme were:

- To strengthen the fundamentals of data science and data-driven problem solving.
- To understand the stages involved in preparing and analysing datasets.

- To develop a systematic approach to interpreting data and drawing meaningful observations.
- To improve practical familiarity with data analysis and machine-learning workflows.
- To connect classroom knowledge with industry-oriented learning through a structured virtual internship.
- To build confidence in presenting technical learning and outcomes in a formal academic report.

4.5 Scope of the Project

The scope of the internship is primarily educational and skill-oriented. It covers the understanding and application of a typical data science workflow, beginning with problem definition and data preparation and continuing through analysis, modelling and interpretation. The internship report documents the learning scope rather than claiming a specific commercial product or medical, financial or other high-stakes deployment.

The knowledge gained during the internship can be applied to future academic projects such as predictive modelling, exploratory data analysis, classification, regression and data visualisation. It also provides a foundation for developing larger machine-learning projects where a complete and reproducible workflow is required.

4.6 Technologies and Tools Used

The internship was focused on data science practice. A typical workflow for this learning domain uses programming, numerical computing, data handling, visualisation and machine-learning tools. The following technologies represent the practical toolset relevant to the internship learning outcomes:

- Python – programming language used widely for data analysis and machine learning.
- NumPy – numerical operations and array-based computation.
- Pandas – dataset loading, cleaning, transformation and tabular analysis.
- Matplotlib – creation of graphs and visual representations of data.
- Scikit-learn – machine-learning workflows, model training and evaluation.
- Jupyter Notebook – interactive environment for combining code, analysis and observations.

4.7 System Architecture

The learning architecture followed a sequential data science workflow. Raw or collected data forms the input layer. The next stage performs data cleaning and preparation, followed by exploratory analysis to identify patterns and relationships. Suitable features are then selected or transformed for modelling. A machine-learning model can be trained on the prepared data and evaluated using appropriate performance measures. Finally, the results are interpreted and communicated through tables, graphs and written observations.

Figure 4.1: General Data Science Workflow

Data Input → Data Cleaning → Exploratory Analysis → Feature Preparation → Model Training → Evaluation → Interpretation

4.8 Methodology

The methodology used for the internship learning process can be represented as a sequence of practical stages. Each stage builds on the previous one and helps develop an understanding of the complete data science lifecycle.

1. **Problem Understanding:** Define the objective, identify the type of data required and determine what kind of output or insight is expected.

2. **Data Collection:** Obtain the relevant dataset from an appropriate source and understand the meaning of each variable.

3. **Data Pre-processing:** Inspect the data, handle missing or inconsistent values, remove unnecessary information and prepare the dataset for analysis.

4. **Exploratory Data Analysis:** Use descriptive statistics and visualisation to understand distributions, relationships, trends and possible anomalies.

5. **Feature Preparation:** Select useful variables and transform the data into a form suitable for the selected analytical or machine-learning method.

6. **Model Development:** Apply a suitable machine-learning approach where prediction or classification is required and train it using the prepared data.

7. **Evaluation:** Assess the model or analytical output using suitable evaluation measures and compare the results against the objective.

8. **Interpretation and Reporting:** Convert technical findings into clear observations and document the process, limitations and outcomes.

4.9 Expected Outcomes

The internship is expected to strengthen practical understanding of the data science lifecycle and improve the ability to work systematically with datasets. It also develops skills in interpreting results, selecting suitable analytical approaches and communicating technical findings in a structured manner.

- Improved understanding of data preparation and exploratory analysis.
- Better familiarity with the relationship between data, features and model performance.
- Improved ability to interpret graphs, statistics and machine-learning outputs.

- Greater confidence in documenting technical work and presenting findings.
- A stronger foundation for future academic projects and industry-oriented data science work.

4.10 CERTIFICATES OF COMPLETION AND COMMUNICATION PROOF

The internship completion certificate is attached in Chapter 3 of this report. According to the certificate, AYUSH KUMAR SINGH of IILM University, Greater Noida successfully completed the 8-weeks Data Science Master Virtual Internship during June-August 2026. The certificate identifies the programme as an EduSkills virtual internship and states that it was supported by SIEMENS.

The certificate also carries an AICTE–EduSkills Virtual Internship 2026 emblem and displays the logos of the Ministry of Education, AICTE, National Internship Portal and EduSkills. The certificate records the certificate ID and student ID for verification.



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Ref No: C17Y2026M071483462

Internship Offer Letter

Student Name	: Ayush Singh
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: June 2026 - August 2026
AICTE Student ID	: NA

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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CHAPTER 5

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[2] Ministry of Education, Government of India – information and resources related to digital and skill development initiatives.

[3] AICTE – resources related to technical education and internship opportunities.

[4] National Internship Portal – internship and student development resources.

[5] Python Software Foundation, Python Documentation – reference material for Python programming and data handling.

[6] Pandas Documentation – reference material for data manipulation and analysis.

[7] NumPy Documentation – reference material for numerical computing.

[8] Matplotlib Documentation – reference material for data visualisation.

[9] Scikit -learn Documentation – reference material for machine-learning algorithms, preprocessing and evaluation.